David Ming

978-501-3858 | dhming@umass.edu | davidming1161@gmail.com | Portfolio

Education

University of Massachusetts Amherst — Amherst, MA
Bachelor of Science in Mechanical Engineering — *2024 – 2028*
GPA: 3.5

Skills

Programming Languages: Python, MATLAB
Tools: SolidWorks, Onshape, PrusaSlicer, Moldex3D

Experience

BSC Technology — Dec 2024 – Jan 2025
Mechanical Engineering Intern — *Shenzhen, China*

- Developed 3D models and contributed to the prototyping process, translating design concepts into tangible product iterations
- Gained hands on factory floor experience in an injection molding and assembly environment, engaging with the full production life cycle from raw material to the finished part.
- Reviewed BOMs and prepared DFM documentation to communicate design feedback to clients

Projects

Cycloidal Drive — 2026 – Present

- Designed a custom cycloidal drive with a 20:1 reduction ratio with a goal of 1,000N/cm
- Modeled the full assembly in SolidWorks, including the eccentric disc geometry
- Selected off the shelf ball bearings for the ring gear rollers and output supports, sized to the drive's load and tolerance requirements

5-DOF Robotic Arm — 2026

- Designed a fully custom multijoint robotic arm from scratch using SolidWorks
- Developed control software to manage joint movement and motion sequencing
- Compatible with an Xbox controller to control movement

FPV Drone — 2023

- Assembled a fully functional FPV drone from components including frame, flight controller, motors, and propellers
- Wired and soldered electrical components, ensuring clean and reliable connections
- Designed and 3D printed custom parts to ensure proper fit and integration between components